Hima — QA Release Testing SOP & Roadmap

Standard operating procedure for testers: what to test **before** release and **after** release

Innovfix Pvt. Ltd. · Hima App · v1 · 2026-07-14 · Owner: Yuvanesh (Tech Lead)

How to use this doc. For every release: run **Phase 1–4 (before release)** and only ship when the sign-off gate passes; then run **Phase 5 (after release)** on production. BEFORE = pre-release on staging/APK. AFTER = post-release on live prod. MUST-PASS = a failure blocks release.

1. Testing Phases (the process)

Phase	When	What
0 · Build sanity	New APK arrives	Installs, opens, no crash on launch, correct version number, login screen loads.
1 · Regression	Before release	Every module's core flows still work (Section 3 checklists).
2 · New-feature	Before release	Deep-test whatever changed in THIS release (from the branch/changelog).
3 · Critical-path	Before release	Money + calls deep test (Section 4) — zero tolerance.
4 · Sign-off gate	Before release	No Sev-1/Sev-2 open. Section 6 checklist signed.
5 · Post-release smoke + monitor	Right after release	Prod smoke on 2–3 real devices + watch live metrics for 2–4h (Section 5).
6 · Rollback watch	First 24h	If a rollback trigger hits (Section 5), revert/hotfix immediately.

Environments & test accounts (set up once)

- **Devices:** ≥2 Android phones (1 low-end, 1 high-end), on WiFi **and** mobile data. Test the **Play Store / Internal Testing** build, not only sideload (sideload can pass while Play build fails — always confirm on the real distribution channel).
- **Accounts:** a **male caller** (with test coins), a **female creator** (verified, with bank/UPI + PAN), and a **fresh new user** (for onboarding). Keep a second creator for block/DND pairing tests.
- **Money:** use small real amounts (₹1–₹49) for real gateway tests — never assume; a coin credit must be verified end-to-end.

2. Bug severity & the release gate

Severity	Examples	Blocks release?
Sev-1 Critical	Coins credited without payment / free coins · double billing · double payout · app crash on launch · calls don't connect · can't log in · money lost	🔴 **YES — hard block**
Sev-2 Major	A gateway fails · withdrawal stuck · matcher connect-rate drops · KYC/verify broken · chat not delivering	🔴 **YES**
Sev-3 Minor	UI glitch, wrong text, non-blocking edge case	🟡 **Ship with a follow-up ticket**

3. BEFORE RELEASE — module-by-module roadmap BEFORE

3.1 Login · Auth · Onboarding

✓	Test
☐	New user: enter mobile → OTP received → verify → profile setup → lands on home.
☐	Wrong OTP rejected; resend OTP works; OTP expiry handled.
☐	Existing user login + logout + re-login (token refresh) works.
☐	Profile update: name (one-time name-change rule), avatar, interests (optional), language.

✓	Test
☐	Gender selection correct; male vs female home screens differ correctly.
☐	AI onboarding chat triggers for new male users; icebreaker shows.
☐	Force-update: a below-min-version user is prompted/blocked per `individual_app_update`.
☐	Delete account flow works; deleted user can't log back in.
☐	**NEW** Keyboard visibility: after entering the mobile number, the **mobile number field is fully visible** above the keyboard (not hidden); and on the OTP step the **6-digit OTP boxes are fully visible** above the keyboard. Both steps, low-end + high-end device. *(QA #4)*

3.2 Creator verification / KYC

✓	Test
☐	Voice verification: record voice → female auto-approve / male-voice reject.
☐	PAN KYC (Cashfree Secure-ID, PaySprint fallback): valid PAN passes, invalid rejected.
☐	Bank + UPI details save + validate; can't withdraw before verification.

3.3 Calls (audio + video) MUST-PASS parts

✓	Test
☐	**Random match:** caller taps random → matched to an online creator in his language → rings.
☐	**Connect:** creator answers → Agora audio/video connects both ways → timer starts (`started_time`).
☐	**Billing:** coins deduct per minute from caller; creator income accrues correctly; amounts match the rate.
☐	**No double-billing** (rapid reconnects / weak network don't charge twice).
☐	**Coins exhaust:** server force-ends the call when caller's coins hit 0.
☐	**Direct call** to a specific creator; recent/favourite call works.
☐	**Reject:** creator rejects → caller sees "busy/unavailable"; 3-reject cool-down applies.
☐	**Missed call:** unanswered → missed_call count + push to creator.
☐	**Video & audio both**; switch between; low-network behaviour (call doesn't hang forever).
☐	Call ends cleanly on either side; end reason recorded; earnings shown to creator.
☐	**NEW** **Network drop while connecting:** caller's network dies mid-ring → the **creator's ring auto-ends** within ~45s (no connected-but-silent call that has to be hung up manually). *(QA #2)*
☐	**NEW** **Ring stops on cancel — app closed:** creator's app is closed/backgrounded and rings → caller cancels → the ringing/vibration **stops immediately** (no ghost ring to dismiss by hand). *(QA #6)*
☐	**NEW** **Call history label:** a connected + completed call always shows its **duration** (even a very short call) in Call History — it is **never labelled "Missed"**. Genuinely unanswered calls still show "Missed". *(QA #10)*
☐	**NEW** **No duplicate/ghost calls on reopen:** after the creator closes and reopens the app, previously received incoming/missed calls do **not** re-appear or re-ring as new incoming calls (queued pushes for already-ended calls are dropped). *(QA #7)*

3.4 Do-Not-Disturb & Blocking

✓	Test
☐	Creator turns DND on → she gets NO incoming calls/chat pings → shows unavailable → stays on until she turns it off.
☐	Call-block: blocked caller can't call her at all; matcher never pairs them again.
☐	Chat-block: blocked caller can't message; existing chat stops.
☐	Both honoured silently (no "you are blocked" popup) across call + chat + matcher.
☐	**NEW** After blocking a contact, that conversation **moves to the bottom** of the chat list (and shows the Blocked badge). *(QA #8)*

3.5 Coins & Payments MUST-PASS

Zero tolerance: coins must be credited ONLY after a verified real payment, exactly once. Test EACH live gateway with a small real amount.

✓	Test
☐	**Cashfree:** buy a pack → pay → coins credited exactly once; check `transactions.add_coins`.
☐	**PhonePe:** same end-to-end.
☐	**Razorpay / UPI link:** same end-to-end.
☐	**Payment FAIL / cancel:** NO coins credited; balance unchanged; clean error to user.
☐	**Webhook replay / forged:** a repeated or unverified webhook does NOT re-credit or free-credit coins.
☐	Offers, trial coins, coupons apply the correct discounted price + coins.
☐	Gateway switch/maintenance flag: when a gateway is off, the app falls back correctly.

3.6 Autopay / Subscription

✓	Test
☐	₹1 trial sheet shows to never-subscribed; mandate created (UPI, GPay-compatible).
☐	Daily coin claim works; recurring ₹299 charge fires; subscription status correct.
☐	Cancel subscription works; CHARGE-fail vs AUTH-fail push messages correct.

3.7 Withdrawals / Payouts MUST-PASS

Zero tolerance: a creator must be paid the correct amount exactly once — never double-paid, never a phantom deduction.

✓	Test
☐	Request withdrawal → balance deducts → Cashfree payout (IMPS) → status Pending→Paid → money reaches bank.
☐	**No double-pay** (rapid taps / retries don't pay twice).
☐	Failed/rejected transfer → balance rolled back correctly.
☐	PAN gate + TDS applied where required; withdrawal blocked without PAN.
☐	Min-amount, bank-detail-missing, and reversal (auto-refund coins) edge cases.

3.8 Creator earnings, tiers & bonuses

✓	Test
☐	Income from calls shows correctly on creator dashboard; balance + `total_income` match.
☐	Tier/score updates; call-bonus tiers pay the right bonus; icebreaker prompts show.
☐	Creator warnings issue + auto-disable + auto-unblock work; star-creator rates apply.
☐	NEW **Call-duration match:** the creator's **Call History duration = Earnings/Transactions duration = the real connected talk time** (measured answer→hangup, with no ring/setup time included). *(QA #12)*
☐	NEW **Every connected call is credited:** a call that connected (`started_time` set) credits the creator for its real talk time **even if the caller's app didn't cleanly report the end** (force-close / network drop) — no answered call is left with ₹0 income. *(QA #11)*

3.9 Chat & Social (real-time)

✓	Test
☐	Send/receive message in real time (both directions); single→blue ticks (read receipts).
☐	Reactions add/update; message delete (archived server-side) + reflects for both.

✓	Test
☐	Bad-word message filtered/handled; presence/typing indicators show.
☐	Attachments upload; gifts send + deduct coins + credit creator; friend requests flow.
☐	Open-chat-from-notification deep-links to the right thread.
☐	NEW Image and audio **attachment options are removed** from the chat composer (emoji available in their place). *(QA #3)*

3.10 Notifications

✓	Test
☐	Push received (FCM + OneSignal): incoming call, new message, friend request, creator-online.
☐	Tapping a notification opens the correct screen; DND suppresses pings.
☐	Notification click/conversion tracking records (if part of this release).
☐	NEW Tapping a **video**-call notification starts a **video** call (not audio) when both call types are enabled. *(QA #9)*
☐	NEW Rejected calls are **not** shown/pushed as "missed"; missed-call notification text shows the correct call status (not both audio & video). *(QA #5)*

3.11 Growth, Games & misc

✓	Test
☐	Install referrer / tracking link / deep link attribute correctly (if changed).
☐	IPL rooms create/join/leave/react; Ludo invite + play (if in release).
☐	Reports/ratings/reviews; app-store rating prompt triggers per rules.

3.12 Admin panel (backend) BEFORE

✓	Test
☐	NEW Block a user/creator with a reason → reopen the account → the **block reason and blocked status/icon are still shown** (retained for audit). *(QA #1)*

4. CRITICAL MUST-PASS scenarios (money + calls) MUST-PASS

These four must be re-verified **every release**, end-to-end, on the real Play build. A single failure = **do not ship**.

1. **Pay → get coins, once.** Real payment on each live gateway → exact coins credited once. Fail/forged → no credit.
2. **Call → correct billing.** Connect → per-minute deduct + creator income correct → no double-bill → force-end on 0 coins.
3. **Withdraw → paid once.** Payout reaches bank, correct amount, no double-pay, rollback on failure.
4. **Matcher + safety.** Random call connects (~normal rate), and DND/blocked pairs are never matched.

5. AFTER RELEASE — production verification & monitoring AFTER

5.1 Smoke test on prod (first 30 min, 2–3 real devices)

✓	Test on the LIVE app
☐	Fresh install from Play → open → login/OTP works.
☐	One real random call connects both ways.
☐	One real small coin purchase credits coins.
☐	One real chat message delivers + blue ticks.

✓	Test on the LIVE app
☐	A test withdrawal request is accepted (status pending).

5.2 Watch these live metrics for 2–4 hours

Metric	Healthy	Rollback trigger
Connect rate (started_time)	~45–52%	Sudden drop >10 pts vs same slot yesterday
Revenue (add_coins/hr)	≈ prior day same hour	Sharp fall vs baseline (not a demand dip)
Gateway success %	Cashfree ~56%, PhonePe ~51%	Any gateway drops >10 pts
Payout %	~40%	Spikes (income leak / ghost calls)
Crashlytics crash-free	>99%	New crash cluster after release
Error log (laravel.log)	baseline noise	New error spiking (e.g. class-not-found, 500s)

> **Version-aware check:** compare the NEW version's users (by `current_version`) vs the stable version on connect rate, ARPU, and gateway success — a bad release shows as the new version underperforming (like v1109 did). Roll out in staged % and watch before 100%.

5.3 Rollback / hotfix criteria

- Any **Sev-1** in production (money loss, free coins, double-pay, mass crash, calls not connecting) → **halt rollout + hotfix/rollback immediately.**
- Connect rate or revenue drops sharply and it's traced to the release (not external) → roll back the staged %.
- A gateway or payout breaks → flip its kill-switch (`gateway_config`) and fix.

6. Sign-off checklist (gate before release)

✓	Item
☐	Build sanity passed (installs, opens, correct version, no launch crash) — on the **Play/Internal** build.
☐	All 4 **MUST-PASS** critical scenarios passed end-to-end.
☐	Regression (Section 3) done for all modules; new-feature deep-tested.
☐	**Zero open Sev-1 / Sev-2.** All logged in the bug tracker with severity.
☐	Tested on ≥2 devices, WiFi + mobile data.
☐	Post-release smoke + monitoring plan assigned (who watches, for how long).
☐	Rollback owner + kill-switches identified. Signed by QA lead + Tech lead.

5. AFTER RELEASE — production verification & monitoring AFTER

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☐	Tested on ≥2 devices, WiFi + mobile data.
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☐	Rollback owner + kill-switches identified. Signed by QA lead + Tech lead.

RELEASE TEST APPENDIX ACX

Local Call-Audio Extraction Foundation

SOP revision 1.1 - 2026-07-17 - Applies to development APK before any transcription provider is connected.

Release gate

ACX-001, 002, 003, 004, 005, 007 and 008 are MUST-PASS on real devices. A compile/unit-test pass alone does not authorize production rollout. Attach logs, profiler screenshots and defect/retest evidence.

ACX-001 - Male local microphone extraction	MUST-PASS / P0
Preconditions / build / role	Development APK; two real devices; male caller and female creator; call connects normally.
Test data	Speak a fixed 65-second script on the male device; female remains quiet.
Steps	1. Start an audio call. 2. Speak before and across 30 s and 60 s boundaries. 3. End normally. 4. Capture filtered AudioExtractor logs.
Expected result	Call audio remains clear. Male device reports monotonic chunks with 30 s max and 1 s overlap after the first. Female local track does not contain male playback. No file/network/AI/warning appears.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

ACX-002 - Female local microphone extraction and attribution	MUST-PASS / P0
Preconditions / build / role	Same two-device build; connected audio call.
Test data	Female speaks fixed script while male stays quiet; then both alternate turns.
Steps	1. Record female-only speech for 35 s. 2. Alternate speakers. 3. Compare each device's local metadata and audible call.
Expected result	Female device marks local speech. Male device marks only its own spoken turns, not remote playback. No speaker-attribution crossover or call-audio regression.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

ACX-003 - Manual mute privacy boundary	MUST-PASS / P0
Preconditions / build / role	Connected call; metadata logs available.
Test data	Speak A while live, mute and speak PRIVATE, unmute and speak B.
Steps	1. Speak A. 2. Tap mute. 3. Speak for at least 10 s while muted. 4. Unmute and speak B. 5. End call.
Expected result	Peer hears A and B only. Pre-mute partial closes; no extracted duration advances during mute; post-unmute starts a clean non-overlapping boundary. PRIVATE is never present in any future payload.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

ACX-004 - Phone/audio-focus interruption	MUST-PASS / P0
Preconditions / build / role	Connected call; second incoming cellular or VoIP/audio-focus event available.
Test data	Speak before, during and after the interruption.
Steps	1. Trigger interruption. 2. Speak while HIMA is held. 3. End interruption. 4. Resume HIMA speech. 5. Repeat once.
Expected result	Extraction pauses with outgoing mic, resumes with a clean boundary, sequence remains monotonic, call does not crash, duplicate workers are not created and held/private speech is excluded.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

Long-Call, VAD and Failure Coverage

Environment: development APK, two real devices, Wi-Fi and mobile data, wired/Bluetooth where available.

ACX-005 - Boundary and hour-long stability	MUST-PASS / P0
Preconditions / build / role	Two charged test accounts; devices on power; Android Studio profiler or equivalent.
Test data	At least 65 min call; scripted words spanning each sampled 30 s boundary.
Steps	1. Run call for 65 min. 2. Sample logs at start, 30 min and 60 min. 3. Exercise speaker, wired and Bluetooth routes. 4. End normally.
Expected result	No 30 s recording stop, missing tail, duplicate-only final chunk, ANR, OOM or increasing worker count. Windows stay <=960 KB; sequence/timestamps are monotonic; audible call quality remains normal.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

ACX-006 - VAD shadow labels across languages	P1
Preconditions / build / role	Connected call in a quiet room, then normal background noise.
Test data	English plus all 11 HIMA languages and code-switching; silence; soft speech; loud speech; fan noise.
Steps	1. Speak each sample naturally. 2. Include quiet and noisy periods. 3. Compare hasSpeech/voiced ratio to the known speaking periods.
Expected result	VAD is language-independent and labels ordinary speech without modifying/cutting PCM. Silence is normally false. Noise false positives are documented; any speech false negative blocks using VAD to skip future API calls.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

ACX-007 - Teardown, reconnect and permission failure	MUST-PASS / P0
Preconditions / build / role	Connected call; ability to toggle network and revoke RECORD_AUDIO.
Test data	Wi-Fi/mobile switch, temporary loss, peer leave, local hangup, permission revoke.
Steps	1. Reconnect network twice. 2. End via peer and local paths. 3. Repeat with permission revoke. 4. Start a fresh call after each case.
Expected result	Observer is not duplicated on reconnect; dispose is safe from leaveChannel/onDestroy; no extractor thread survives teardown; next call captures normally; existing call failure handling remains authoritative.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

ACX-008 - Privacy, no-spend and regression gate	MUST-PASS / P0
Preconditions / build / role	Fresh development APK; proxy/file inspection tools; admin access for read-only verification.
Test data	One 2-minute call containing harmless test speech.
Steps	1. Inspect app private cache/database before/during/after. 2. Inspect network requests. 3. Check admin/warnings. 4. Verify call settlement/rating flow.
Expected result	Zero audio files, uploads, transcript rows, provider calls, AI cost, admin audio, moderation results or warnings. Existing connection, mute, timer, hangup, settlement and rating flows still complete.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RELEASE TEST APPENDIX CTN

Creator-online Notification Call-Type Routing

SOP revision 1.2 - 2026-07-17 - Production payload hotfix. Applies to both instant and scheduled OneSignal creator-online senders.

Release gate	CTN-001, CTN-002 and CTN-003 are MUST-PASS. Do not send a production test notification without owner approval. Record payload evidence, device evidence, defect ID and retest status.

CTN-001 - Instant video notification routes to video

MUST-PASS / P0

Preconditions / build / role	Backend hotfix on all three nodes; instant flag ON; current production APK; male caller and verified female creator; creator has both audio and video enabled.
Test data	Use an approved test creator/caller pair. Enable video last so the instant alert advertises video.
Numbered steps	1. Capture the generated OneSignal data object without exposing tokens. 2. Confirm call_type=video. 3. Tap the alert on the male device. 4. Observe the connecting and in-call screen.
Expected result	Payload contains source=creator_online, the intended creator ID and call_type=video. Tap opens the video connecting flow and joins a video call; it never starts audio.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

CTN-002 - Scheduled fallback preserves video type

MUST-PASS / P0

Preconditions / build / role	QA/demo environment or an explicitly approved controlled production test; instant path disabled or deliberately made to leave the queue; scheduled creator processor active; both creator call types enabled.
Test data	Queue one creator-online video event for the approved pair.
Numbered steps	1. Let creators:process-notifications claim the queue row. 2. Capture redacted payload evidence. 3. Confirm call_type=video. 4. Tap the resulting alert.
Expected result	Scheduled payload and human text agree on video. Tap opens video. The fallback no longer loses call_type or silently selects audio.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

CTN-003 - Audio notification remains audio

MUST-PASS / P0

Preconditions / build / role	Current APK; creator has both types enabled; approved instant and scheduled test paths available.
Test data	Enable audio last so the advertised type is audio.
Numbered steps	1. Exercise instant sender and scheduled sender separately. 2. Verify call_type=audio in each redacted payload. 3. Tap each notification.
Expected result	Both alerts open the audio connecting flow. Video routing fix does not invert or break existing audio behavior.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

CTN-004 - Availability and stale-type fallback	P1
Preconditions / build / role	Current APK; controlled accounts; ability to change creator audio/video availability before tap.
Test data	Video-only, audio-only, both-off, and a payload whose advertised type becomes unavailable before tap.
Numbered steps	1. Test video-only and audio-only alerts. 2. Turn the advertised type off before tapping while leaving the other type on. 3. Test both-off. 4. Record resulting screen/message.
Expected result	Available advertised type is honored. If it became unavailable, app uses the remaining available type. Both-off does not launch a billable call and shows the existing unavailable handling.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

CTN-005 - Legacy payload compatibility	P1
Preconditions / build / role	Debug/QA injection only; current APK; do not send a real provider notification.
Test data	Creator-online payload omitting call_type, representing an old queued/provider payload.
Numbered steps	1. Inject/open the legacy payload in QA. 2. Test audio-only, video-only and both-enabled creator states. 3. Confirm app remains stable and clears stored notification routing keys after use.
Expected result	No crash. Audio-only and video-only still open their available type; both-enabled retains the documented backward-compatible audio fallback only for legacy payloads.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

CTN-006 - Production smoke and regression evidence	P1 / AFTER
Preconditions / build / role	Approved production smoke window; all three live hashes match; PHP-FPM 8.2/8.4 active; owner approval for any outward test notification.
Test data	One approved audio alert and one approved video alert; no customer accounts or PII in evidence.
Numbered steps	1. Verify live source/hash and service health. 2. Send only the approved test alerts. 3. Tap each on a current APK. 4. Monitor Laravel/PHP errors and creator-online send failure rate for 30 minutes. 5. Record rollback decision.
Expected result	Audio opens audio and video opens video. No new PHP errors, send failures, duplicate sends, latency regression, database work or availability regression. Reverse the two payload lines if a rollback trigger occurs.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RELEASE TEST APPENDIX RNG

Incoming Ring Stops After Caller Disconnect

SOP revision 1.3 - 2026-07-17 - Authenticated Redis heartbeat plus legacy fallback. No user_calls migration and no call-row write from liveness polling.

Release gate	RNG-001 through RNG-005 are MUST-PASS before rollout. Run network, Redis-failure and load cases in QA. Production calls or notifications require separate owner approval. Record build, timing, redacted evidence, defect and retest status.

RNG-001 - Original ghost-ring reproduction on legacy APK

MUST-PASS / P0

Preconditions / build / role	QA backend patch; production-equivalent APK that does not send call_ring_heartbeat; caller and recipient test accounts; incoming screen polling active.
Test data	Audio and video, both call directions, caller network disabled during pre-answer setup.
Numbered steps	1. Start the call. 2. Confirm recipient is ringing. 3. Disable caller network before Answer. 4. Leave recipient untouched. 5. Measure from call creation and from network loss. 6. Repeat with caller app killed.
Expected result	Recipient ringtone, Telecom call and incoming notification all stop by 47.5 seconds from creation. No Answer screen remains and no old ring reappears. No coins are deducted because started_time remains empty.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RNG-002 - Authenticated heartbeat stops a disconnected caller quickly

MUST-PASS / P0

Preconditions / build / role	QA backend patch; new APK containing authenticated heartbeat; valid JWT; clocks synchronized for evidence only.
Test data	Drop Wi-Fi, mobile data, airplane mode and process kill separately at 5-10 seconds of ringing.
Numbered steps	1. Start ringing and capture successful heartbeat responses. 2. Trigger one disconnect mode. 3. Measure last accepted heartbeat to recipient teardown. 4. Repeat each mode for audio/video and both caller genders.
Expected result	Recipient teardown occurs in 15-18 seconds after the last accepted heartbeat. check_call_alive returns alive=false, ringable=false and reason=not_answered. The call row is not changed by the liveness poll.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RNG-003 - Answer and heartbeat timeout boundary

MUST-PASS / P0

Preconditions / build / role	QA backend patch and new APK; controllable latency or packet loss; both devices online initially.
Test data	Answer attempts immediately before and after the 15-second stale boundary, including a delayed heartbeat response.
Numbered steps	1. Delay caller heartbeat traffic. 2. Tap Answer at 14, 15 and 16 seconds in separate calls. 3. Observe accept-in-flight behavior, Agora join, server response and billing state. 4. Repeat 20 times.
Expected result	An Answer already in flight is not dismissed by a late poll callback. A call declared unreachable does not open a billable empty room. No row contains contradictory started and terminal state; no double ring occurs.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RNG-004 - Heartbeat authorization and ownership	MUST-PASS / P0
Preconditions / build / role	API test client in QA; one male-initiated and one female-initiated call; caller, recipient and unrelated JWTs.
Test data	Missing token, invalid token, recipient token, unrelated token, caller token, invalid and nonexistent call IDs.
Numbered steps	1. POST each credential/call combination. 2. Verify status and response. 3. Inspect Redis keys without exposing tokens. 4. Confirm no user_calls write occurred.
Expected result	Only the actual initiator can refresh the heartbeat. Missing/invalid authentication is rejected, recipient/unrelated users receive forbidden, malformed IDs are rejected, and no database row is mutated.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RNG-005 - Delayed push, recovery and no resurrection	MUST-PASS / P0
Preconditions / build / role	QA backend patch; new APK; ability to delay FCM delivery and background/foreground the recipient app.
Test data	Fresh ring, heartbeat-stale ring, 30-second age-guard ring and 45-second legacy-terminal ring.
Numbered steps	1. Delay each incoming push. 2. Trigger foreground pending-call recovery. 3. Deliver the old push after each boundary. 4. Reopen the app. 5. Attempt a new unrelated call.
Expected result	Only a fresh unanswered call is surfaced. ringable=false blocks stale/started calls, pending recovery does not resurrect them, and a later new call rings normally.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RNG-006 - Redis failure and backward-compatible fallback	P1
Preconditions / build / role	QA only; controlled Redis unavailability or injected Cache exception; legacy and new APKs.
Test data	Cache read failure, write failure, recovery during ring and heartbeat endpoint timeout.
Numbered steps	1. Inject each failure. 2. Start a call and attempt heartbeats. 3. Observe API status, logs and recipient ring. 4. Restore Redis and repeat.
Expected result	Cache failure does not end a legitimate call early and does not return a server 500. The new client degrades to the 45-second legacy terminal response; recovery needs no database repair.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RNG-007 - Capacity and database regression gate	P1 / LOAD
Preconditions / build / role	Staging or isolated production-equivalent environment; representative concurrency; query and Redis telemetry.
Test data	At least 100 concurrent ringing sessions for five minutes plus current check_call_alive traffic.
Numbered steps	1. Generate 2.5-second authenticated heartbeats and both-side liveness polls. 2. Record p50/p95/p99 latency, Redis ops, PHP CPU/RSS and DB queries. 3. Verify query plan. 4. Compare baseline.
Expected result	Heartbeat is bounded to about 24 requests/minute per ringing caller, uses one caller-auth lookup plus one user_calls PK lookup and one Redis SET, creates no migration or call-row write, and does not materially regress API/DB capacity.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

RNG-008 - Three-node production smoke and rollback	P1 / AFTER
Preconditions / build / role	Separate owner approval for node patch, PHP-FPM reload and any real-device call; all three node hashes verified first.
Test data	One approved legacy-client call and one approved new-client call; no customer data in evidence.
Numbered steps	1. Verify route/service/controller hashes on app-1/2/3. 2. Verify PHP-FPM 8.2/8.4. 3. Run approved calls. 4. Monitor 4xx/5xx, Redis latency and PHP errors for 30 minutes. 5. Record rollback decision.
Expected result	All nodes behave identically, ordinary calls connect normally, legacy teardown meets 47.5 seconds, new teardown meets 15-18 seconds, and no billing, availability, database-write or service-health regression appears.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

Call Moderation Compact Controls

Demo admin UI revision - 2026-07-17 - Blade-only presentation change; control routes and behavior remain unchanged.

CMR-UI-001 - Desktop one-line control rail	MUST-PASS / P0
Preconditions / build / role	Demo admin account with Call Moderation access; desktop viewport at least 1200px; moderation settings visible.
Test data	Master On and Off states, auto-warning On and Off states, minimum version 0 and a positive version code.
Numbered steps	1. Open Call Moderation Review. 2. Inspect the first control area at 1200px and 1920px widths. 3. Verify labels, values and state colors. 4. Expand Recent changes.
Expected result	Moderation, Auto warning and Minimum app version appear in one compact row. Each control has a short label, current value and Save button. No former large headings or explanatory paragraphs remain. Recent changes stays available below the row without clipping or overlap.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

CMR-UI-002 - Control behavior and authorization regression	MUST-PASS / P0
Preconditions / build / role	Authorized demo admin; separate unauthorized or insufficient-role session; known initial control values; no real call required.
Test data	Toggle each switch once and restore it; version values 0, valid positive integer, negative value and value above 2147483647.
Numbered steps	1. Change each control and press its Save button. 2. Confirm the warning prompt. 3. Reload and verify persistence/history. 4. Test invalid version boundaries. 5. Repeat update requests without authorization.
Expected result	Existing PATCH routes, CSRF, confirmations, permission checks, validation and audit history still work. Invalid values are rejected. Unauthorized users cannot update controls. Master, auto-warning and version semantics are unchanged, and all test values are restored after execution.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

CMR-UI-003 - Responsive, keyboard and error-state QA	P1
Preconditions / build / role	Desktop and mobile browser/device; keyboard-only input; browser zoom 100% and 200%; ability to simulate validation error and infrastructure-disabled state.
Test data	Viewports 390px, 768px, 900px and 1200px; long translated/admin font rendering where available.
Numbered steps	1. Resize through each viewport. 2. Tab through switches, input, Save and Recent changes. 3. Test at 200% zoom. 4. Trigger a validation error. 5. Inspect disabled controls.
Expected result	The controls remain one line on desktop and stack cleanly below 900px. Focus is visible, labels are announced, touch targets remain usable, errors remain visible, disabled controls cannot submit, and the evidence locator and call list retain their prior layout and behavior.
Execution	Status: NOT RUN | Evidence: ____________________ | Defect: __________ | Retest: __________

Hima — Support Bot

AI ticket-deflection feature — **how it works**, configuration, and operations

Innovfix Pvt. Ltd. · Hima App · v1 · 2026-07-17 · Backend: Laravel (demo_hima / hima-admin-panel) · App: Android (hima_app)

> **What it is, in one line.** A guided in-app chat that sits **in front of** the existing ticket system: it asks the user 3–4 questions, reads **their own live account data**, answers, and only creates a support ticket if the user says the answer did not help. The goal is to stop the repeat-contact tickets (the average user raised ~2 per month) at the source — without ever making a false statement about someone's money or account.

1. The user flow

Step	Screen	Notes
0 · Language	"Choose your language" (prompt in English, all 11 options)	Applies to **this conversation only** — does not change the profile language / matching.
1 · Category	5 categories, volume-ordered	Calls · Earnings/Withdrawal · Coins/Recharge · Account/Profile · Something else.
2 · Sub-category	The category's sub-options + "My issue isn't listed"	Chips are **server-driven** — the app renders labels and posts keys; it knows nothing about the taxonomy.
3 · One detail	A sub-category-specific question (optional, with Skip where safe)	e.g. "double charge" asks for a date/time; the answer genuinely needs it.
4 · Answer	Typing → the AI's answer	Only this step is slow. The bot has already read the user's data before replying.
5 · Did this solve it?	[Yes, solved] · [No, still need help]	**Yes** → close, **no ticket**. **No** → one more attempt, then a human.

> **Yes vs No — the whole point.** "Yes" closes the chat and **creates no ticket row** (a deflected issue must not appear in `tickets`, or deflection is unmeasurable). "No" buys **one** second, different answer; a second "No" creates a ticket for a human, pre-tagged and with the bot's diagnostics attached. There is no third attempt.

2. How it stays accurate (why it can be trusted with money)

- **Reads live data via fixed, read-only tools** — never free-form SQL. Eight tools: account summary, recent calls, call detail, recharge history, coin ledger, withdrawal history, earnings summary, block status.
- **The user is structural, not a parameter.** The logged-in user id is fixed on the server; it appears in **no** tool the model can call, so the model cannot express "look at another user's data" (closes IDOR by construction).
- **Grounding guard.** Every number in a reply (₹ amounts, coins, order ids, times) must be a value a tool actually returned. Any invented number → the answer is discarded and the issue is escalated.
- **Skip-AI on money.** For sensitive/redirect money categories the model is never called at all — a human takes it directly, so money cannot be hallucinated about.
- **Recharge crediting is never asserted as fact.** `transactions` has no gateway/order link and `add_coins` also covers admin grants, so a same-size credit near a payment is only an **indication**, not proof. The bot states **payment_status** (paid / not completed) as fact, reports any nearby credit as a labelled possibility with its confidence, and routes every paid recharge to a human to verify against the gateway dashboard — it never claims "you were credited" or "money taken, no coins".
- **Fail closed.** If a tool cannot read the user's data (DB/replica down), the bot escalates to a human rather than answering around the gap. On prod, reads use a replica; if the replica lags, it falls back to the primary so it never states stale data.
- **Never promises** a refund, credit or timeline; **never deletes** an account — a delete request is treated as a symptom and probed.

3. Configuration & kill-switches (ops)

Setting	Where	Effect
`support_bot_enabled`	`gateway_config`	**Master kill-switch.** When set to `0`, the app shows the old raise-ticket form and in-flight sessions escalate to a human instead of calling the model. Fails **closed** (unreadable = off).
`support_bot_model`	`gateway_config`	The model id (default Claude Sonnet). Deliberately separate from the ticket-answering model.
`support_bot_subcategories`	`gateway_config`	CSV allowlist of sub-categories (empty = all). An off-list topic routes straight to a human — turn the bot on one topic at a time as shadow data proves each playbook.
`known_issue_banner`	`gateway_config`	When set, shown first during a platform incident — deflects the duplicate flood before the menu.
`support_hours_start` / `_end`	`gateway_config`	Agent shift (default 4 PM–12 AM IST) for the out-of-hours "when we'll look" ETA.
`SUPPORT_BOT_READ_CONNECTION`	.env	Read replica for tool queries on prod (empty on demo). Lag-guarded.
`SUPPORT_BOT_SESSION_CEILING`	.env	Hard per-session deadline (25s) — under PHP's execution limit; the app waits longer so it always gets the definitive answer.
`SUPPORT_BOT_TOKEN_BUDGET`	.env	Hard per-session cost ceiling; exceeding it escalates rather than spending more.

> **Answers are ops-editable without a deploy.** The per-sub-category playbooks live in `ai_reference_faqs` (already per-language, already an admin UI). The Yes/No taps are a free labelled dataset — a sub-category below ~40% "Yes" is a broken playbook or a real product bug.

4. Endpoints (all `auth:api`, no user_id ever accepted from the client)

Endpoint	Purpose
`support_bot_start`	Opens/*resumes* a session; returns language chips + any known-issue banner. Checks for a pending escalated ticket up front.
`support_bot_reply`	Walks the tree, then answers. One request per session at a time (locked).
`support_bot_feedback`	"Did this solve it?" — the only thing that resolves or escalates. Runs the second attempt.
`support_bot_session`	Rebuilds a session reopened after the app was killed (transcript + current step).
`support_bot_attach`	Optional file on the ticket the bot raised (image/video/audio, ≤5MB). Serialised per ticket.

5. Rollout & what to watch

- **Demo-first, always.** Everything is verified on `demohima.himaapp.in` before prod. Prod goes on an explicit go, staged by sub-category allowlist and user %.
- **Fallback is safe.** With the kill-switch off, or on any start failure, the user gets the old form — the support channel is never removed behind the LLM.
- **Weekly metrics:** per-sub-category Yes-rate · grounding-guard trips (should be ~0) · p50/p95 latency · tokens/session · **48h re-contact rate** · and the north star: **tickets per user per month.**

> **Design guardrail — "calls not coming" (the #1 driver) is a real product bug the bot cannot fix.** The bot reads the data and escalates; it does **not** troubleshoot ("check your network"), which reads as being fobbed off by a robot. Its deflection target there is near-zero **by design** — track it as a product-bug signal, not a deflection KPI.

6. Technical inventory

Controllers, services & models

Component	Type	Responsibility
`SupportBotController`	Controller	The 5 endpoints; step machine; session lock; kill-switch guard; escalation + attach.
`SupportBotService`	Service	The model tool-loop; deadline budget; grounding guard; escalate control-flow.
`SupportToolRunner`	Service	The 8 read-only tools; user-scoped by constructor; replica-lag guard; recharge matcher.
`TicketCreationService`	Service	Turns a session into a human ticket; per-user advisory lock; canonical-phrase routing.
`SupportBotSafetyGate`	Service	Safety-only refusal (harassment/abuse/minors/self-harm/report).
`SupportHours · SupportBotStrings · SupportBotTaxonomy · Prompts\SupportBotPrompt`	Service	Shift/out-of-hours ETA · 11-language strings · categories & canonical phrases · the versioned system prompt.
`SupportBotSession`	Model	Per-conversation state (step, language, transcript, last_render, telemetry).
`Ticket · Users`	Model	Existing models; bot writes tickets with `source=bot`; reads Users for scope.

Tables (reads unless noted)

Table	Use
`support_bot_sessions` (RW)	Session state + telemetry. New table; migrations are `hasColumn`-guarded.
`tickets` (RW)	Human tickets on escalation (bot columns: source, sub_category, language, bot_session_id, screenshot).
`gateway_config`	Kill-switch, model id, known-issue banner, support hours. Runtime, cross-node.
`users · user_calls · transactions · coins · cashfree_payments · phonepe_payments · blocked_users`	Read-only tool sources (account, calls, ledger, packs, payments, block status).
`ai_reference_faqs`	Per-language answer playbooks (admin-editable, no deploy).

External providers · jobs

- **OpenRouter → Claude Sonnet** (via `gateway_config.support_bot_model`) — the only external call; synchronous, budgeted (25s ceiling, token cap), keyed by `openrouter_api_key`.
- **Cashfree / PhonePe — read-only**, via their payment tables; the bot never calls the gateways.
- **Jobs/cron:** the bot adds **none** — every request is synchronous. Escalated tickets flow through the existing ticket pipeline. No queue worker to run or monitor.

Failure modes & handling (all fail toward a human, never a wrong answer)

Failure	Handling
Model timeout / API error / round limit	Escalate to a human ticket (deadline budget → `escalate`).
Tool can't read data (DB/replica down)	Fail closed — escalate (`tool_error`), never answer around the gap.
Replica lagging	Read from the primary for that session.
Ungrounded number in the reply	Discard the answer, escalate (grounding guard).
Recharge crediting unprovable	Report payment_status only; route to a human to verify (no credited/no-coins claim).
Kill-switch off (start or mid-session)	Old raise-ticket form / escalate the in-flight session; no model call.
Duplicate ticket race / concurrent attach	Per-user & per-ticket MySQL advisory locks (cross-node).
Node-local attachment storage	**Known operational risk** — an upload may be unavailable from another node until ticket storage is moved to a shared disk (platform task).

Rollback

- **Instant:** set `gateway_config.support_bot_enabled=0` — users get the old form, in-flight sessions escalate. No deploy, cross-node.

- **All users, staged by topic:** the bot is on for every user when enabled (no per-user gating, by owner decision). Turn it on one topic at a time with `support_bot_subcategories` (a `gateway_config` allowlist; empty = all topics) — off-list topics route straight to a human.
- **Schema:** all new columns are additive and `hasColumn` -guarded; reverting the branch leaves the tables harmless. No destructive migration.

7. QA test cases (formal — run before release)

> **How to run.** Each case has a stable ID (`SB-###`), priority (P1 blocks release), module reference, preconditions and test data, numbered steps, and the expected result. Fill the **Execution** row: mark Pass/Fail, attach Evidence (screen recording / DB row), log a Defect ID on failure, and record the Retest result. **Build/role/data:** run on the Play/Internal build, as a **male caller** and a **female creator**, on WiFi **and** mobile data, unless a case says otherwise.

`SB-001` P1 **Guided flow, end to end** Module: Support Bot / Flow · Ref §1

Precondition	Bot enabled (`support_bot_enabled=1`); test account has no open escalated ticket.
Test data	Any account; pick category "Calls".
Steps	1. Open Help → Support. 2. Observe the language prompt; select one. 3. Select a category, then a sub-category. 4. Answer the follow-up (or Skip if offered). 5. Wait for the answer, then observe the "Did this solve it?" controls.
Expected	Steps appear in order language → category → sub-category → optional detail → answer → Yes/No. "My issue isn't listed" is present at the sub-category step. No crash; each step responds <2s except the answer.

Execution: □ Pass □ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

`SB-002` P1 **Language & script consistency** Module: Support Bot / Language · Ref §1.0

Test data	Select English at step 0; then type the free-text detail in Tanglish ("calls varala").
Steps	1. Select a language and complete the flow to an answer. 2. Read every bot message for language consistency. 3. At the free-text step type romanised (Tanglish) text.
Expected	The whole conversation stays in the chosen language (no mid-chat switch). The answer mirrors the user's script (Tanglish in → Tanglish out), not native Tamil.

Execution: □ Pass □ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

`SB-003` P1 **Deflection creates no ticket** Module: Support Bot / Deflection · Ref §1.5

Precondition	DB/admin access to check the `tickets` table.
Steps	1. Complete a flow to an answer. 2. Tap "Yes, solved". 3. Query `tickets` for this user/timestamp.
Expected	Conversation closes; optional rating shows; **no new row** in `tickets` .

Execution: □ Pass □ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

`SB-004` P1 **Escalation: second attempt then human** Module: Support Bot / Escalation · Ref §1.5

Steps	1. Reach an answer; tap "No, still need help". 2. Observe the second attempt (a different answer). 3. Tap "No" again. 4. Open the ticket in admin.
Expected	Exactly one second attempt (different content). A ticket is created for a human (`source=bot` , `requires_manual_support=1`) with the bot's diagnostics attached. No third AI attempt.

Execution: □ Pass □ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

SB-005 P1 **Delete-account is probed, never actioned** Module: Support Bot / Safety · Ref §2

Steps
1. Choose Account → a delete-account intent (or type "delete my account").
2. Follow the conversation.

Expected The bot asks what the underlying problem is and tries to help; it never offers or performs deletion. No account is deleted or disabled.

Execution: □ Pass □ Fail **Evidence:** ___________ **Defect#:** _____ **Retest:** _____

SB-006 P1 **Resume mid-flow after app kill** Module: Support Bot / Resume · Ref §4

Steps
1. Advance to the sub-category / detail step.
2. Force-kill the app (swipe from recents).
3. Reopen Help → Support.

Expected The prior conversation is rebuilt and lands on the same step; the user does not re-answer earlier questions.

Execution: □ Pass □ Fail **Evidence:** ___________ **Defect#:** _____ **Retest:** _____

SB-007 P2 **Resume after answer / when finished** Module: Support Bot / Resume · Ref §4

Steps
1. Receive an answer (Yes/No shown); kill and reopen.
2. Resolve or escalate the session; kill and reopen again.

Expected First reopen shows THAT answer with Yes/No (never the rejected first answer). After resolve/escalate, reopen is read-only (transcript only, no live input).

Execution: □ Pass □ Fail **Evidence:** ___________ **Defect#:** _____ **Retest:** _____

SB-008 P2 **Attachment — validation & lifecycle** Module: Support Bot / Attach · Ref §4

Precondition A ticket has just been raised (SB-004), so the attach option is offered.

Test data A <5MB image; a >5MB video; a .txt file; a large screen-recording.

Steps
1. Attach the valid image → confirm it appears on the ticket.
2. Attach the >5MB file → observe rejection.
3. Attach the .txt → observe type rejection.
4. Start a large upload, then rotate the screen; and separately, start one and background/kill the app.
5. Try to pick a second file while one is uploading.

Expected Valid file uploads (visible on the ticket in admin). Too-large/wrong-type are rejected with a clear message and are NOT shown as accepted. Rotation mid-upload does not break the upload or crash. Killing mid-upload leaves no orphaned cache file. A second pick while uploading is blocked with a "wait" message.

Execution: □ Pass □ Fail **Evidence:** ___________ **Defect#:** _____ **Retest:** _____

SB-009 P2 **Slow-network patience** Module: Support Bot / Timeout · Ref §1.4

Test data Throttle the network (e.g. slow 3G profile).

Steps
1. Reach the answer step on a slow connection.
2. Repeat for the second attempt (after "No").

Expected The app waits (up to ~30s) for the definitive answer or escalation on BOTH the first and second attempts; it does not fail early or dump the user to the old form.

Execution: □ Pass □ Fail **Evidence:** ___________ **Defect#:** _____ **Retest:** _____

SB-010 P1 **Kill-switch (start & mid-session)** Module: Support Bot / Ops · Ref §3

Precondition Ability to set `gateway_config.support_bot_enabled`.

Steps
1. Set the flag to 0; open Help → Support.
2. Set it back to 1; start a session and reach the sub-category step.
3. Set it to 0 mid-session; send the next step.

Expected With the flag 0, the old raise-ticket form is shown (no bot). Flipping to 0 mid-session makes the next step escalate to a human (no AI call), with no hang or crash.

Execution: □ Pass □ Fail **Evidence:** ___________ **Defect#:** _____ **Retest:** _____

SB-011 P2 **Pending ticket & duplicate guard** Module: Support Bot / Tickets · Ref §1

Precondition	The test account already has one open escalated ticket.
Steps	1. Open Help → Support. 2. Separately, from two devices, drive one session each to escalation at the same time.
Expected	The user is told about the open ticket at the START (not after four questions). No duplicate ticket is created, even from two concurrent devices.

Execution: ☐ Pass ☐ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

SB-012 P1 **MONEY — recharge is verified, never invented** Module: Support Bot / Money · Ref §2 · MUST-PASS

Precondition	DB access; a creator and a male account. Verify every figure against the DB, never the reply alone.
Test data	A real ₹1–₹49 recharge; also one completed >30 min after starting the order.
Steps	1. Ask a recharge/coins question after a real recharge. 2. Compare the bot/ticket diagnostics to the `cashfree_payments` / `phonepe_payments` and `transactions` rows. 3. Repeat for a delayed (>30 min) completion and for two close same-size purchases.
Expected	Recharge routes to a human. Diagnostics state **payment_status** as fact only; crediting is shown as INDICATIVE (never a definitive "you were credited" or "money taken, no coins"). A delayed credit is still matched; two close same-size purchases with one credit are marked for human review, not falsely credited. Every ₹/coin/order-id/time in any reply came from the data (no invented refund/timeline).

Execution: ☐ Pass ☐ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

SB-013 P1 **MONEY — earnings connected count** Module: Support Bot / Money · Ref §2 · MUST-PASS

Precondition	A creator account with some calls that never connected (`started_time='00:00:00'`).
Steps	1. Ask an earnings question. 2. Compare the "connected calls" figure to the Recent Calls tool and the DB.
Expected	The connected count excludes `00:00:00` (never-connected) calls and matches Recent Calls exactly.

Execution: ☐ Pass ☐ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

SB-014 P1 **SECURITY — own data only (IDOR)** Module: Support Bot / Security · Ref §4 · MUST-PASS

Precondition	Two accounts A and B; a tool to send raw API requests (Postman).
Steps	1. As A, drive a bot session; confirm it only ever reflects A's data. 2. With A's token, call the ticket routes passing B's user_id / mobile number.
Expected	The bot only ever shows A's own data. Ticket create/list use the token identity — passing B's user_id/mobile returns A's own tickets or 401, never B's data; unregistered vs registered numbers return identical responses (no enumeration).

Execution: ☐ Pass ☐ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

SB-015 P3 **Zoho launcher hygiene** Module: Support Bot / UI · Ref §5

Steps	1. As a female account (launcher enabled from home), open Support and leave. 2. As a male account, open Support and leave.
Expected	Female: the floating Zoho launcher is hidden on the bot screen and restored on exit. Male: the launcher never appears.

Execution: ☐ Pass ☐ Fail **Evidence:** ____________ **Defect#:** ______ **Retest:** ______

Support Bot Selected-Language Authority

Bug fix SBLANG-2026-07-20. These MUST-PASS cases supersede SB-002 wherever SB-002 says typed text can change the reply language. The chatbot selection now always wins.

SB-016 | P1 MUST-PASS | Original reproduction and multi-turn regression

Module / ref	Support Bot / Language / Multi-turn - SBLANG-2026-07-20 - prompt v10
Build / environment	Play/Internal Android build connected to demo backend with the support bot enabled. Run on WiFi and mobile data.
Account role	Female creator whose app profile language is Tamil. Repeat once with a male caller.
Preconditions	No open escalated ticket. Start a new bot session. Do not reuse the affected terminal session from 2026-07-20.
Test data	Inside the chatbot select Kannada. Choose Calls & Connection, then Call auto-cut / disconnects. Enter Last user. When asked for a time, answer in English or Tamil text.
Steps	1. Open Help and Support and start a new chatbot session. 2. Select Kannada even though the account profile is Tamil. 3. Complete the Calls & Connection / disconnect path using the test data above. 4. Read the category prompt, detail prompt, AI clarification and final AI answer. 5. Reply to the clarification in English or Tamil and continue to the final answer. 6. Background and reopen the app; verify the restored transcript and active input state. 7. Tap No once and inspect the second AI attempt. 8. Capture screen recording plus the redacted session language, prompt_version and step from admin/DB evidence.
Expected	- Every user-visible bot line after selection is Kannada. The Tamil profile, English chip label and Tamil/English free text never switch the reply language. - Clarification uses a text box, the reply is accepted, and the final/second answer remains Kannada. - The restored transcript does not re-render a Tamil answer or reopen the wrong step. - The session stores language=Kannada and prompt_version=v10-20260720. No users.language update occurs. - No duplicate provider request, duplicate answer or duplicate ticket occurs on background/resume.
Execution	[ ] Pass [ ] Fail Evidence: ____________________ Defect ID: __________ Retest: __________

Release rule

A single Tamil, English or other-language AI bubble after Kannada is selected is a release blocker. Do not accept a correct script with the wrong spoken language (for example Romanized Tamil).

Support Bot Selected-Language Authority

Cross-language matrix and fail-closed behavior. Provider output must be stubbed for the negative checks; do not spend or transmit real user data merely to force a bad model response.

SB-017 | P1 MUST-PASS | All-language matrix and wrong-language fail-safe

Module / ref	Support Bot / Language Guard - SBLANG-2026-07-20 - all 12 chatbot languages
Build / environment	Backend unit/integration environment with a fake OpenRouter response, plus one normal demo smoke per selected language. No live provider fault injection.
Account role	At least two test accounts. Pair each selected language with a different profile language.
Test matrix	English, Tamil, Hindi, Telugu, Kannada, Malayalam, Bengali, Marathi, Gujarati, Odia, Punjabi and Assamese. Test native-script and Romanized output for the 11 Indian languages; test clear standard English in Latin script for English.
Steps	1. For each language, set the account profile to a different language and select the target only inside the chatbot. 2. Complete one normal category and a multi-turn clarification; verify fixed strings and AI text stay in the selected language. 3. In the fake-provider test, return a confident different Romanized language (including the reported Tamil response for selected Kannada). 4. Verify one corrective rewrite requests only the selected language in a verifiable script: standard English/Latin for English and native script for the 11 Indian languages. 5. Make the fake provider return the wrong language again and observe the terminal behavior. 6. For selected English, inject confident Romanized Tamil and verify it is blocked; then return clear English and verify it passes. 7. Inspect the model-visible account summary and tool schema for absence of users.language. 8. Repeat with an unsupported/empty session language and verify no provider/tool call begins. 9. Record request count, response time, ticket count and sanitized language-guard logs.
Expected	- The selected chat language is the sole authority in all 12 rows; profile, typed text, FAQ/tool content and prior turns cannot override it. - English is a guarded support-chat choice: clear English passes, while confidently detected Romanized Indian-language output is rejected. - A valid selected-language answer passes without an extra provider call. - A wrong or unverifiable answer gets at most one bounded rewrite. The wrong text is never displayed. - If the rewrite is still wrong, the flow fails closed to one human ticket and uses fixed translated scaffolding in the selected language. - An invalid session language fails before tool/provider work. No profile-language fallback occurs. - Normal API/query count is unchanged; guard CPU remains negligible and no unbounded retries are possible.
Regression	Re-run SB-001, SB-004, SB-005, SB-006, SB-007, SB-009, SB-010, SB-011 and SB-014. Confirm grounding, timeout, idempotency, resume, ticket and kill-switch behavior are unchanged.
Execution	[ ] Pass [ ] Fail Evidence: ____________________ Defect ID: __________ Retest: __________